

Prognostic Role of Physical Function in Older Adults with Acute Myeloid Leukemia Treated by Lower-intensity Therapy: A Prospective Cohort Study

Running title: Physical Function in AML on Lower-Intensity Therapy

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Research in context

Evidence before this study

Previously, Klepin et al. showed in 2013 that geriatric assessment (GA) can predict survival in older adults with acute myeloid leukemia (AML) receiving induction chemotherapy. Our recent prospective cohort study proved that measuring physical function with the Short Physical Performance Battery (SPPB) and depression using the short form of the Geriatric Depression Scale could predict survival in older adults with AML who received intensive chemotherapy. However, as venetoclax (VEN)-based therapy gains wider usage in older adults with AML, evaluating the prognostic value of GA in this therapeutic regimen becomes necessary, yet limited data exists on this topic. In particular, validating the prognostic relevance of objectively measured physical function in a lower-intensity setting is necessary, as it has already been established in the intensive chemotherapy setting. We searched PubMed without language and publication date restrictions using the following search terms: "physical function" AND "acute myeloid leukemia" and did not find any clinical trials or retrospective studies that met our needs. We also expanded our search to include "geriatric assessment" AND "acute myeloid leukemia". And then we found that Ritchie et al. investigated the utility of GA in older adults receiving lower-intensity therapy using data from the CALGB trial and found prognostic associations with comorbidity burden, low quality of life, and cognitive dysfunction. However, the patients in their study received decitabine and bortezomib, not VEN-based therapy, and they did not use objective measures of physical function. To determine if there is a risk model including physical function terms for older adults with AML receiving less intensive treatment, we searched PubMed using the search terms "prediction" AND "acute myeloid leukemia" AND "older". We found the most recent model by Sasaki et al. in 2023, based on a retrospective analysis of older adult AML patients receiving lower-intensity treatment, but they did not include GA variables in the model.

Added value of this study

In the new era of VEN, our paper is a timely and important study that provides new data to support the use of SPPB to identify older patients who are more likely to tolerate and benefit from hypomethylating agents (HMA)/VEN. Although HMA/VEN therapy has demonstrated superior survival benefits compared to HMA monotherapy, there are also concerns about the higher toxicity of HMA/VEN, such as persistent cytopenia. Categorization of patients at risk for HMA/VEN therapy is a critical need at this time.

Implications of all the available evidence

Our study highlights that objective measures of physical function can predict survival in older adults with AML who are receiving HMA/VEN therapy, but not in patients with HMA monotherapy. This suggests that physical function has a different effect on prognosis depending on the type of treatment. Furthermore, our proposed prognostic models incorporating objective physical function into preexisting models would help to select patients who will benefit from HMA/VEN and require an adjusted dosing schedule in this vulnerable population. Our findings also suggest that assessment of physical function should be done before assigning participants to clinical trials that compare intensive chemotherapy with the HMA/VEN combination in AML patients who are considered fit. As a future direction, it is necessary to see how changes in GA domains during the course of treatment correlate with patient outcomes. It may also be desirable to evaluate cognitive function in patients receiving HMA/VEN to determine prognostic associations. Finally, our study will facilitate the development and testing of interventions to improve physical function in these vulnerable patients.

74 Summary

75 **Background:** The Short Physical Performance Battery (SPPB) and the short form of the
76 Geriatric Depression Scale (SGDS-K) predict survival outcomes in older adults with acute
77 myeloid leukemia (AML) treated with intensive chemotherapy. However, their role in lower-
78 intensity therapy remains unclear. We evaluated their prognostic role in older adults with AML
79 treated with hypomethylating agents (HMA) with or without venetoclax (VEN).

80 **Methods:** This single-center, prospective cohort study was conducted at Seoul St. Mary's
81 Hospital, Seoul, Republic of Korea. AML patients, aged at least 60 years, receiving HMA or
82 HMA/VEN therapy were included. SPPB and SGDS-K scores were assessed before treatment,
83 and their associations with overall survival (OS) or other outcomes, and prognostic utility were
84 analyzed. SPPB and SGDS-K assessments and outcomes were collected manually by the
85 researchers.

86 **Findings:** From November 2016 to June 2023, 124 (HMA: $n = 52$, HMA/VEN: $n = 72$) patients
87 were enrolled. Their data were collected until 1 January 2024. Impaired physical function
88 (SPPB < 10) was significantly associated with worse OS in patients receiving HMA/VEN
89 (median 5.4 months [2.5-9.4] vs. 16.4 months [8.7-not reached], $p = 0.0059$), but not in HMA-
90 treated patients (median 7.8 months [3.3-17.7] vs. 8.6 months [2.4-22.8], $p = 0.9195$). The
91 inferior survival in the HMA/VEN group with physical impairment was attributed to higher
92 rates of death within 60 days (impaired vs. non-impaired: 29.4% vs. 2.6%, $p = 0.0021$).
93 Impairment on the SGDS-K showed no association with OS in either group. In the HMA/VEN
94 group, incorporation of the SPPB into existing prediction models for older adults with AML
95 (Wheatley index and treatment-related mortality score model) improved their predictability of
96 early death.

Interpretation: This study highlights the prognostic role of objectively measured physical function on survival in older adults with AML treated with HMA/VEN therapy, suggesting that a treatment strategy may need to be established based on this finding.

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102 Introduction

103 The median age of patients diagnosed with acute myeloid leukemia (AML) is 68 years,
104 with approximately one-third of patients diagnosed at 75 years or older.¹ Older adults with
105 AML have a significantly worse prognosis than younger patients their treatment options have
106 been limited, with one- and five-year survival rates of 26·3% and 7·8%, respectively.² Recently,
107 combination therapy with hypomethylating agent (HMA) and venetoclax (VEN) has
108 demonstrated superior outcomes compared to HMA monotherapy,³⁻⁵ which became the first
109 recommended therapy in patients with AML unfit for intensive chemotherapy.⁶ Given the new
110 therapeutic options with varying intensities, there is a huge clinical need for decision-making
111 tools to determine the optimal treatment regimen for older adults with AML. Several prognostic
112 models have been developed to help identify patients at high risk of either early death (ED) or
113 treatment resistance.⁷⁻⁹ However, these models include a limited number of patients who
114 received treatments that included VEN, and they need recalibration to account for new
115 therapies and changes in supportive care.¹⁰

116 There is a general consensus that multidimensional geriatric assessment (GA) tools should
117 be used to evaluate the mental and physical functions of older patients to measure their
118 treatment fitness instead of relying solely on laboratory profiles.¹¹ Klepin et al. previously
119 showed that GA can predict survival in older adults with AML receiving induction
120 chemotherapy,¹² and we also have shown in a prospective cohort study that assessing physical
121 function using the Short Physical Performance Battery (SPPB) and depression using the short
122 form of the Geriatric Depression Scale (Korean version, SGDS-K) could predict survival
123 outcomes in older adults with AML treated with intensive chemotherapy.¹³ However, there is
124 no data whether the SPPB and SGDS-K scores are prognostic in patients treated with HMA
125 with or without VEN. In this context, we report the results of a single-institution prospective

cohort study to evaluate the prognostic role of physical function scores and depression scales in older adults with AML treated with lower-intensity therapy, HMA, or HMA/VEN.

Methods

Study design and patients

Initially, we conducted a single-center, prospective cohort study, enrolling adult patients newly diagnosed with AML between November 2016 and December 2019. The study included individuals aged ≥ 60 years who underwent intensive induction chemotherapy or lower-intensity therapy and completed GA before the initiation of treatment (registered in the Clinical Research Information Service, CRIS, #KCT0002172, Cohort #1). Since VEN was introduced in Republic of Korea in March 2020 and has become the main treatment for older patients, we recognized the need for a new cohort study including patients treated with VEN. Therefore, we conducted another cohort study (CRIS #KCT0002261, Cohort #2) between January 2020 and June 2023. For Cohort #2, we decided to perform only two GA measures: the SPPB and SGDS-K, which were significantly associated with patient outcome in the Cohort #1 study. Finally, the current study evaluated patients from Cohort #1 and Cohort #2 who received lower-intensity therapy, either HMA or HMA/VEN. Their data were collected until January 1, 2024. Figure 1 describes the patient enrollment process. The Institutional Review Board of the Catholic Medical Center approved this study. The analyses were conducted in accordance with the guidelines of the Institutional Review Board and followed the ethical principles outlined in the Declaration of Helsinki.

Treatment and geriatric assessment

HMA monotherapy consisted of administering 20 mg/m² intravenous decitabine (DEC) for five days or 75 mg/m² subcutaneous azacitidine (AZA) for seven days. The HMA/VEN

treatment protocol was the same as the pivotal VIALE-A trial protocol and our previous study comparing DEC/VEN with DEC monotherapy.^{3,4} Cyto reduction of white blood cell (WBC) counts to $<20 \times 10^9/L$ using hydroxyurea, low-dose intravenous cytarabine, or leukapheresis was performed before both treatments. Fluconazole (400 mg/day) was administered as antifungal prophylaxis during the neutropenic period in all patients, and no antibiotics were used for antibacterial prophylaxis. Both treatments were maintained until disease refractoriness was confirmed or until the patient could tolerate the therapy. If the patient became eligible for hematopoietic stem cell transplantation (HSCT) and consented, we proceeded with HSCT.

The SPPB consists of three tests: the gait speed test (measuring the time taken to walk 4 m), sit-and-stand speed test (measuring the time taken to stand up from a chair and sit down 5 times), and balance tests (including the side-by-side stand, semi-tandem stand, and tandem stand, each held for 10 s). Each measurement was scored from 0 to 4, with a higher score indicating better performance.¹⁴ The SGDS-K is a screening tool used to assess depression in elderly populations. It consists of 15 survey items, with each item scored 1. Higher scores on the SGDS-K indicate a higher risk of depression.¹⁵ According to the development groups, physical impairment based on the SPPB is categorized as follows: severe (less than 4 points), moderate (4-6 points), mild (7-9 points), and minimal (10 or more points).¹⁶ For the SGDS-K, a score of 8 or higher was classified as high-risk for depression.¹⁵ In addition to the SPPB and SGDS-K, patients in Cohort #1 were evaluated the full geriatric assessment domains as per the Cohort #1 study design, and the methods of each test are documented in our previous study.¹³

Outcomes

The primary endpoint was overall survival (OS), defined as the time from the start of treatment to death from any cause or the date of last follow-up. Secondary outcomes included event-free survival (EFS), according to the definition proposed by the European LeukemiaNet (ELN) in 2022,¹⁷ and treatment responses (initial and best responses). The ELN 2022 criteria

is used to determine complete response (CR), CR with partial hematologic recovery (CRh), CR with incomplete hematologic recovery (CRi), and other responses.¹⁷ Composite CR is a combination of CR, CRh, and CRi. We also analyzed the rates of ED (death within 60 days of treatment), and cumulative incidence of death (CID) (in composite CR patients), following the ELN 2022 definition.¹⁷

Statistical analysis

Variables are summarized as medians with interquartile ranges (IQRs) or counts with percentages. The Wilcoxon rank-sum and chi-squared tests (Fisher's exact test for low-frequency counts) were used to compare continuous and categorical variables, respectively, and Kaplan-Meier estimates were used to assess survival outcomes, which were compared by the log-rank test. Cox models were constructed to assess the prognostic significance of GA with adjustment for the effect of covariates, including age, Eastern Cooperative Oncology Group score (ECOG), secondary AML, and ELN 2022 risks. For CID, cumulative incidence estimates, considering relapse as a competing risk, were calculated and compared using Gray's test. To evaluate the utility of GA variables when combined with preexisting models, we computed the Wheatley index, which is commonly used in older adults with AML,⁹ and the treatment-related mortality (TRM) score for AML.^{8,18} We then investigated the impact of incorporating GA variables into the models on the likelihood of models, integrated discrimination improvement (IDI), continuous net reclassification improvement (NRI),¹⁹ multivariate receiver operating characteristic (ROC) curves, and their area under the curves (AUC).²⁰ In addition, we evaluated the benefit of decision making based on the GA variables using the decision curve analysis proposed by Vickers et al.²¹ All analyses were performed using R software (version 4.2.2, R Foundation for Statistical Computing, Vienna, Austria). Statistical significance was set at $p < 0.0500$.

Role of funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

Baseline characteristics

During the study period of two cohorts, 468 older adults with AML were screened, of whom 145 agreed to participate in the cohorts and received lower-intensity treatment. Twenty-one of them were excluded, leaving 124 patients in the final analysis (Figure 1). The median observation period of patients was 7·8 months (IQR: 2·5-15·0 months), and that of survivors was 13·7 months (8·7-26·5 months). Table 1 shows the baseline characteristics of 52 and 72 patients receiving HMA and HMA/VEN, respectively. In both treatment groups, most patients received DEC as HMA. The HMA/VEN group exhibited a significantly lower age distribution than the HMA group (median age: 75 vs. 71 years, $p = 0·0013$), while other characteristics were generally similar. Most patients in both HMA and HMA/VEN treatment groups fell into the poor prognostic and high-risk groups based on the Wheatley index (71·2% and 65·3%, respectively) and the TRM score model (75·0% and 59·7%, respectively). Among the enrolled patients, 15·3% of the HMA/VEN group underwent HSCT, which was significantly higher than that of the HMA group, in which only one patient underwent HSCT ($p = 0·0137$). Baseline characteristics of HMA patients in Cohort #1 and Cohort #2 were not significantly different (data not shown).

The median time from diagnosis to the SPPB and SGDS-K assessment was 8 days (range, 3-12 days). Lower-intensity treatment was started within three days after the SPPB and SGDS-K measurements were completed. Before the SGDS-K evaluation, 3 patients had been under regular consultation with a neuropsychiatric department, with two having a major depressive

disorder and the other having a seizure disorder. The total SPPB score was significantly lower in the HMA group than in the HMA/VEN group (median 8 vs. 10, $p = 0.0009$). The SGDS-K score showed a more adverse trend in the HMA group than in the HMA/VEN group (median 4 vs. 3, $p = 0.0397$). Details are summarized in Supplementary Figure 1. GA results other than SPPB, and SGDS-K in Cohort #1 patients are summarized in Supplementary Table 1.

SPPB, SGDS-K, and survival outcomes

During the observation period, Forty-six (88.5%) HMA patients and 44 (61.1%) HMA/VEN patients died. We compared the OS of HMA patients between Cohort #1 and Cohort #2 to see if the difference in OS between two cohorts was significant, and there was no significant difference in the OS (data not shown). In the HMA group, there was no significant difference in OS according to the presence of physical impairment using a cut-off of 10 points for the SPPB (Figure 2A). In contrast, in the HMA/VEN group, significantly poor survival was observed in patients with impaired SPPB compared to those without impairment (Figure 2B), and hazard ratio for OS by SPPB impairment was 2.29 (1.25-4.18, likelihood ratio test $p = 0.0092$). the trends observed in OS were attenuated in the regards of EFS because patients who did not achieve a composite CR by the defined landmark (180 days) or died before the defined landmark without response assessments were calculated as a day-1 event (Figure 2C and 2D). In the HMA/VEN group, the association between the individual test domains of the SPPB and OS also significantly impacted survival (Supplementary Figure 2 and 3).

In both HMA and HMA/VEN group, there was no significant difference in OS according to the risk of impairment in the SGDS-K (Supplementary Figure 4A and 4B). When we further investigated the association between depression risk and the need for neuropsychiatric interventions during the treatment, including psychiatric counseling and medication prescriptions, considering death a competing risk (Supplementary Figure 4C and 4D), patients at a high risk of depression underwent neuropsychiatric intervention at a significantly higher

rate than those who were not at increased risk in the HMA/VEN group (70·0% vs. 51·0%, $p = 0·0243$).

In multivariate Cox models incorporating SPPB and SGDS-K (as grouped variables) with covariates (Supplementary Figure 5A and 5B), we observed that the SPPB score served as a significant independent predictor of OS in the HMA/VEN group (HR for SPPB impairment group: 2·23 [1·17-4·23], $p = 0·0143$), but not in the HMA group. The SGDS-K scores did not demonstrate any significant association with any of the models.

SPPB and treatment response, ED, and CID

The treatment responses of the patients are summarized in Table 2. There were no significant differences in the treatment response based on physical impairment in either treatment group. The difference in ED rates based on physical impairment was not significant in the HMA group (Table 2 and Supplementary Figure 6A), whereas in the HMA/VEN group, there was a significant difference in ED rates according to physical impairment, with 29·4% of impaired patients dying within 60 days compared to 2·6% of non-impaired patients (Fisher's exact test $p = 0·0021$) (Table 2 and Supplementary Figure 6B). The progress of ED patients is summarized in Supplementary Figure 6C. There was one patient each who died early after achieving CRi, CRh, and MLFS in the HMA/VEN group, and the remaining patients died before response assessment or were unresponsive to treatment. Most early deaths were caused by pneumonia. On the other hand, in patients with confirmed composite CR, CID showed a similar trend regardless of SPPB impairment in both HMA and HMA/VEN groups. (Supplementary Figure 7).

Improvement of existing survival prediction models by SPPB

Based on our findings so far, we have focused on identifying the implications of SPPB for the HMA/VEN group. we evaluated whether incorporating the SPPB score into the

Wheatley index and TRM score in HMA/VEN group enhanced the model's performance in predicting ED (Figure 3). The Wheatley index significantly improved upon including the SPPB tests (AUC: 0.6803 to 0.8033, $p = 0.0306$), so as on the TRM score (AUC: 0.6835 to 0.7985, $p = 0.0113$), (Figure 3A and 3B). Additionally, we conducted a decision curve analysis for ED. Namely, the feasibility of the prediction model at different thresholds was examined by calculating the weighted difference between the true positive (death occurred in cases where the model predicted a probability of death above the threshold) and the false positive (where the model predicted a probability of death above the threshold but no death occurred) rates. Decision curve analysis also showed that integrating the SPPB score into the original models of the Wheatley index and TRM score enhanced the net benefit across a wide range of threshold probabilities, encompassing the observed ED rate (15.3% in our data) (Figure 3C and 3D).

Discussion

In this study, we tested the pre-treatment physical function and screened for depression in prospective cohorts of older adults with AML treated with HMA or HMA/VEN. We found that the SPPB score did not have prognostic significance in the HMA monotherapy group, whereas it emerged as a strong prognostic factor in the HMA/VEN group. A SPPB score was significantly associated with ED. Even after adjusting for covariates, the SPPB score remained an independent predictor of OS in the HMA/VEN group. Including the SPPB score improved the performance of existing survival prediction models, such as the Wheatley index or TRM score, particularly in predicting ED. In contrast, we did not observe a survival difference based on the depression scale in this lower-intensity therapy cohort.

The different prognostic impacts of objectively measured physical function using the SPPB between the HMA and HMA/VEN groups can be attributed to several factors. First, the expected survival benefit with HMA is modest compared to that with HMA/VEN,^{3,4,22,23}

resulting in poor overall survival outcomes for patients in this group, regardless of physical impairment. In contrast, HMA/VEN therapy has demonstrated superior survival benefits compared to HMA monotherapy;^{3,4} however, there are also concerns about the higher toxicity of HMA/VEN, such as persistent cytopenia.²⁴ Therefore, the HMA/VEN group may have exhibited significant differences in ED according to the presence of physical impairment, leading to a difference in OS. Second, the prognostic effect of physical function may be less pronounced in frail individuals. The patients in the HMA group were less fit than those in the HMA/VEN group regarding age, and SPPB distribution. Accordingly, the HMA group had significantly inferior physical function based on the total SPPB score and distribution (minimally limited by the SPPB test: HMA vs. HMA/VEN, 33·3% vs. 57·7%). Further studies with a larger randomly assigned cohort based on age, ECOG findings, or physical function status will answer these questions.

We found that in the impaired physical function group, although the HMA/VEN treatment showed a better response compared to HMA alone (composite CR: HMA vs. HMA/VEN: 13·0% vs. 50·0%), this response did not translate into a survival benefit due to increased ED caused mainly by infectious complications. Adjusting the dosing schedule is recommended to reduce the treatment intensity in this vulnerable population. Aiba et al. recently reported the feasibility of shortening the duration of VEN treatment by demonstrating that a 2-week schedule had similar response rates and lower febrile neutropenia, warranting a large-scale study to confirm this finding.²⁵ Conversely, physically fit patients experienced a strong benefit from HMA/VEN treatment in the current study. These findings suggest that the assessment of physical function should be considered when randomizing participants in clinical trials that compare intensive chemotherapy with the HMA/VEN combination for AML patients considered fit.²⁶

Consistently, logistic regression models and decision curve analyses showed that the SPPB score significantly improved the discrimination of high risk patients and added decision benefit to the Wheatley index and TRM score. Given that the Wheatley index or TRM score models included a wide range of variables, including age, secondary AML, ECOG, cytogenetics, and blood chemistry findings, the increase in model performance with SPPB not only reminds us that physical function has a prognostic impact independent of conventional variables in this patient population, but also points to the feasibility and need for direct clinical application of SPPB. If it is difficult to perform a full SPPB test in practice, a simpler test domain may be useful. We found that the inclusion of only a simpler measure, such as the gait speed test score, could also improve the predictability of early mortality of Wheatly index and TRM score (data not shown).

In contrast to our previous investigations involving patients undergoing intensive chemotherapy,¹³ the current study did not find any correlation between depressive symptom burden and survival or baseline physical function. Given the differences in patient characteristics between the two studies, baseline depressive symptoms may more strongly influence baseline physical function in younger patients with better ECOG performance status.¹³ It is also plausible that the lower intensity and different treatment environments, mainly outpatient treatment of HMA or HMA/VEN, mitigated cancer mortality associated with depression through a pathophysiological effect via neuroendocrine and immunological functions or by weakening treatment adherence.²⁷ In addition, our cohort did not include a sufficient number of patients at high risk of depression, which may have limited our ability to establish a significant relationship between depression risk and survival. The prognostic significance of baseline depressive symptoms and dynamic changes throughout the treatment course should be further evaluated in larger cohorts, particularly those treated with DEC/VEN, where baseline physical function is prognostic.

The strengths of our study include its prospective nature and the scarcity of research on GA in patients with AML treated with lower-intensity therapy. To our knowledge, this is the first prospective study to evaluate the prognostic role of objective physical function measurement in older adults with AML treated with lower-intensity therapy (HMA with or without VEN). Additionally, we demonstrated the extent to which physical function measures improved the predictability of preexisting prognostic models. Nonetheless, the modest size of the cohort and the data from a single institution could limit its generalizability. Another limitation is the potential selection bias caused by the substantial number of patients screened but not enrolled in this study. However, survival outcomes in each treatment group were not significantly different between the enrolled and unenrolled patients (Supplementary Figure 8), suggesting that the bias may not be significant. And the assessment of physical impairment using the SPPB, which focuses on lower extremity strength, may not accurately reflect the true physical function of patients with leg-related disabilities or acute problems localized to the legs. In such cases, tests such as hand grip strength should be considered an alternative measure of physical function.²⁸

As a future direction, it would be intriguing to see how changes in GA domains during the course of treatment correlate with patient outcomes. Saad et al. conducted post-remission GA measurements after induction in older adults with AML and reported that depressive symptom and SPPB scores were significantly associated with mortality.²⁹ A similar study design could be attempted in HMA/VEN patients. And since we did not investigate the GA domains other than SPPB, and SGDS-K in Cohort #2, future research is warranted. Ritchie et al. investigated the correlation between GA measures and prognosis in unfit AML patients over 60 years of age treated with decitabine plus bortezomib.³⁰ They found that cognitive function measures such as the Blessed Orientation-Memory-Concentration score were significantly associated with patient survival. We examined the association of each GA domain with OS in

HMA-treated patients in Cohort #1 who underwent complete GA, and no significant associations were found (Supplementary Table 1). It is desirable to evaluate cognitive function in patients receiving HMA/VEN to determine prognostic associations.

In summary, our study highlights the prognostic role of objectively measured physical function on survival in older adults with AML treated with HMA/VEN therapy, suggesting a different prognostic impact of physical function in the context of treatment type. Moreover, we demonstrated that incorporating the SPPB into existing models for older adults with AML receiving HMA/VEN improved their predictability. Comprehensive prognostic models incorporating appropriate GA domains along with disease and laboratory characteristics would aid in selecting the optimal treatment for older adults with AML.

Authors' contributions

Conception and design: D. Kwag, B.S. Cho

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Analysis and interpretation of data: D. Kwag, B.S. Cho

Writing, review, and/or revision of the manuscript: D. Kwag, B.S. Cho, and H.J. Kim

Administrative, technical, and material support: S. S. Park, J. H. Yoon, and S. E. Lee.

Study supervision: B.S. Cho

Declaration of interest

The authors declare that they have no conflicting interests.

Data sharing

For the original data, please contact the corresponding author (cbscho@catholic.ac.kr)

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486

487 **Figure legends**

488 **Figure 1. Enrollment of patients.** AML: Acute myeloid leukemia. CGA: Comprehensive
 489 geriatric assessment, ECOG: Eastern Cooperative Oncology Group, EF: Ejection fraction,
 490 FEV1: Forced expiratory volume in one second, HMA: Hypomethylating agent, SGDS-K:
 491 Short Form of Geriatric Depression Scale (Korean version), SPPB: Short Physical Performance
 492 Battery, VEN: Venetoclax. *Korean version of Instrumental Activities of DailyLiving (K-
 493 IADL), Korean version of the modified Barthel index (K-MBI), Handgrip strength, SPPB,
 494 Mini-Mental State Examination (MMSE) in the Korean version, Nursing Delirium Symptom
 495 Checklist (KNU-DESC), SGDS-K, Patient Health Questionnaire-9 (PHQ-9), National
 496 Comprehensive Cancer Network's Distress Thermometer (NCCN-DT), Older Americans
 497 Resources and Services (OARS), and Mini Nutritional Assessment (MNA). †Low dose
 498 cytarabine, n = 3; Gilteritinib, n = 1. ‡Low dose cytarabine, n = 2; Clinical trial, n = 6.

499 **Figure 2. A–B. Overall survival by SPPB impairment in patients. C–D. Event-free**
 500 **survival by SPPB impairment in patients.** P-values are calculated from log-rank tests. CI:
 501 Confidence interval, EFS: Event-free survival, HMA: Hypomethylating agent, NR: Not
 502 reached, OS: Overall survival, SPPB: Short Physical Performance Battery, VEN: Venetoclax.

503 **Figure 3. A-B. Changes in discrimination indices by incorporating SPPB into the**
 504 **Wheatley index or TRM score.** Curves are ROC curves for predicting ED. C-D. **Decision**
 505 **curve analysis of original comparing Wheatley index and TRM score with SPPB**
 506 **incorporating models.** AIC: Akaike information criterion, AUC: Area under the curve, ED:
 507 Early death, HMA: Hypomethylating agent, integrated discrimination improvement, NRI: net
 508 reclassification improvement, ROC: Receiver operating characteristic, SPPB: Short Physical
 509 Performance Battery, TRM: Treatment-related mortality, VEN: Venetoclax.

Table 1. Baseline characteristics of patients

Variables (Median with IQR or Frequency with percentage)	HMA (N: 52)	HMA/VEN (N: 72)	<i>p</i>
Decitabine as HMA	47 (90.4)	66 (91.7)	1.0
Age	75 (69-77)	71 (67-74)	0.0013
≥ 75 years	29 (55.8)	16 (22.2)	0.0003
Male	32 (61.5)	41 (56.9)	0.7428
ECOG ≥ 2	17 (32.7)	16 (22.2)	0.2731
Secondary AML	7 (13.5)	11 (15.3)	0.9801
BM blasts, %	56.5 (40-81.5)	53 (34-81.2)	0.4751
WBC, per 10⁹/L	19.9 (3.2-75.5)	14.3 (2.2-50.9)	0.1854
Hemoglobin, g/dL	9.1 (7.4-10.2)	8.1 (7.1-9.4)	0.0272
Platelet, per 10⁹/L	51.0 (33.0-88.2)	53.5 (33.2-89.2)	0.8365
Peripheral blasts, %	23 (7-67.5)	23.5 (8-43.2)	0.8792
Creatinine, mg/dL	0.9 (0.7-1.1)	0.9 (0.7-1.0)	0.2585
Albumin, g/dL	3.8 (3.3-4.0)	3.9 (3.5-4.2)	0.0182
ELN 2022 chromosome			0.1492
Favorable	5 (9.6)	8 (11.1)	
Intermediate	38 (73.1)	41 (56.9)	
Adverse	9 (17.6)	23 (31.9)	
ELN 2022 risk			0.8599
Favorable	11 (21.2)	15 (20.8)	
Intermediate	21 (40.4)	26 (36.1)	
Adverse	20 (38.5)	31 (43.1)	
Mutations			
<i>FLT3</i> -ITD	11 (21.2)	13 (18.1)	0.8410
<i>NPM1</i>	14 (26.9)	21 (29.2)	0.9428
<i>CEBPA</i> (bZIP-inf)	2 (3.8)	0 (0)	0.1739
<i>TP53</i>	5 (9.6)	8 (11.1)	0.7733
<i>RUNX1</i>	9 (17.3)	5 (6.9)	0.0895
<i>ASXL1</i>	6 (11.5)	5 (6.9)	0.5241
Poor Wheatley index	37 (71.2)	47 (65.3)	0.7013
High risk TRM score	39 (75.0)	43 (59.7)	0.1138
Proceed with HSCT	1 (1.9)	11 (15.3)	0.0137

AML, acute myeloid leukemia; ECOG: Eastern Cooperative Oncology Group, ELN: European LeukemiaNet, HMA: Hypomethylating agent; HSCT: Hematopoietic stem cell transplantation; IQR: Interquartile range; TRM: Treatment-related mortality; VEN: Venetoclax, WBC: White blood cell.

Table 2. Chemotherapy response

	HMA			HMA/VEN		
Frequency (%)	SPPB <10 (N = 35)	SPPB ≥10 (N = 17)	<i>p</i>	SPPB <10 (N = 34)	SPPB ≥10 (N = 38)	<i>p</i>
Initial response						
Composite CR	5 (14·3)	4 (23·5)	0·4514	11 (32·4)	13 (34·2)	1·0
CR	4 (11·4)	2 (11·8)		4 (11·8)	3 (7·9)	
CRh	0 (0)	1 (5·9)		1 (2·9)	1 (2·6)	
CRi	1 (2·9)	1 (5·9)		6 (17·6)	9 (23·7)	
MLFS	2 (5·7)	1 (5·9)		7 (20·6)	6 (15·8)	
PR	1 (2·9)	0 (0)		1 (2·9)	0 (0)	
NR	16 (45·7)	8 (47·1)		10 (29·4)	17 (44·7)	
NE	11 (31·4)	4 (23·5)		5 (14·7)	2 (5·3)	
Best response						
Composite CR	13 (37·1)	6 (41·2)	1·0	17 (50·0)	22 (57·9)	0·6641
CR	8 (22·9)	4 (23·5)		6 (17·6)	9 (23·7)	
CRh	1 (2·9)	1 (5·9)		2 (5·9)	3 (7·9)	
CRi	4 (11·4)	1 (5·9)		9 (26·5)	10 (26·3)	
MLFS	0 (0)	1 (5·9)		5 (14·7)	4 (10·5)	
PR	1 (2·9)	0 (0)		0 (0)	0 (0)	
NR	10 (28·6)	6 (35·3)		7 (20·6)	10 (26·3)	
NE	11 (31·4)	4 (23·5)		5 (14·7)	2 (5·3)	
Death in 30 days	4 (11·4)	2 (11·8)	1·0	4 (11·8)	1 (2·6)	0·1817
Death in 60 days	7 (20·0)	5 (29·4)	0·4959	10 (29·4)	1 (2·6)	0·0021

CR, complete response; CRh: Complete response with partial hematologic recovery; CRi: Complete response with incomplete hematologic recovery; HMA: Hypomethylating agent; MLFS: Morphologic leukemia-free state; NE, not evaluated; NR, no response; PR: Partial response; SPPB: Short Physical Performance Battery, VEN: Venetoclax.

Cohort #1: Full CGA* evaluated

Nov 2016 ~ Dec 2019

Screened (N: 212)

Newly diagnosed, older AML

Consent denial (N: 19)

Intensive (N: 10) / HMA (N: 9)

Intensive treatment (N: 125)**Best supportive care (N: 17)****Others[†] (N: 4)****Non-intensive treatment: HMA (N: 47)**

Primary cause of lower-intensity treatment

- Age ≥ 75 (N: 23)
- ECOG ≥ 2 at treatment (N: 7)
- Adverse karyotype (N: 5)
- Adverse molecular genetics (N: 4)
- Decreased EF/FEV1 at baseline (N: 3)
- Patient refusal of intensive treatment (N: 2)
- Other physician's decision (N: 3)

Excluded (N: 8)

- Not in a condition to perform physical tests : HMA (N: 5)
- Not in a condition to read and answer the questionnaire : HMA (N: 3)

Enrolled (N: 39)

HMA (N: 39)

Cohort #2: SPPB, SGDS-K evaluated

Jan 2020 ~ Jun 2023

Screened (N: 256)

Newly diagnosed, older AML

Consent denial (N: 24)Intensive (N: 7) / HMA (N: 3) /
HMA/VEN (N: 14)**Intensive treatment (N: 107)****Best supportive care (N: 19)****Others[‡] (N: 8)****Non-intensive treatment: HMA (N: 20) / HMA/VEN (N: 78)**

Primary cause of lower-intensity treatment

- Age ≥ 75 (N: 22)
- ECOG ≥ 2 at treatment (N: 11)
- Adverse karyotype (N: 14)
- Adverse molecular genetics (N: 5)
- Decreased EF/FEV1 at baseline (N: 21)
- Patient refusal of intensive treatment (N: 13)
- Other physician's decision (N: 12)

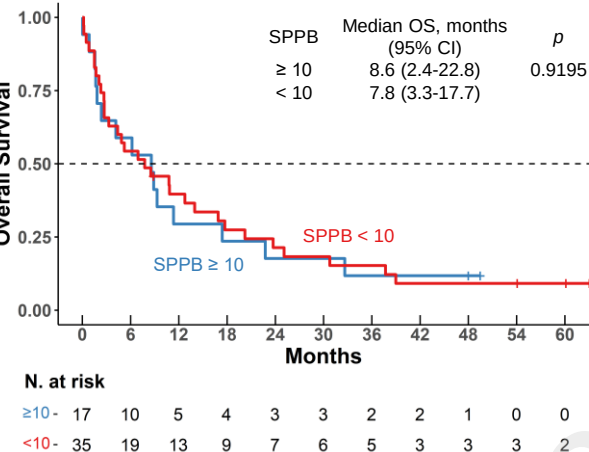
Excluded (N: 13)

- Not in a condition to perform physical tests : HMA (N: 4) / HMA/VEN (N: 2)
- Not in a condition to read and answer the questionnaire: HMA (N: 3) / HMA/VEN (N: 4)

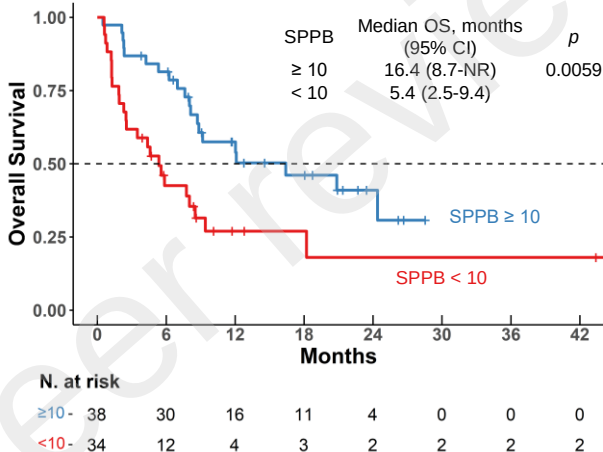
Enrolled (N: 85)

HMA (N: 13) / HMA/VEN (N: 72)

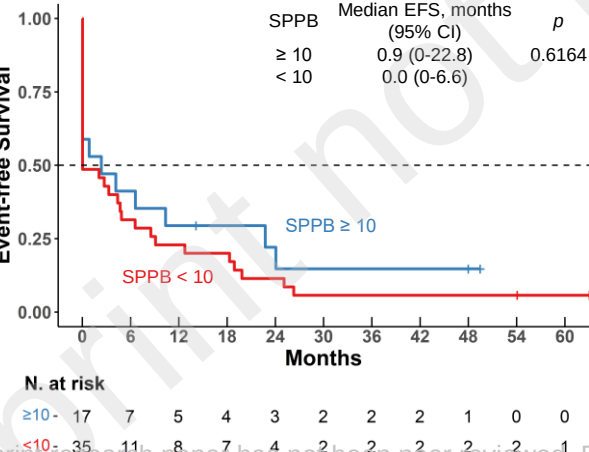
A. OS - HMA



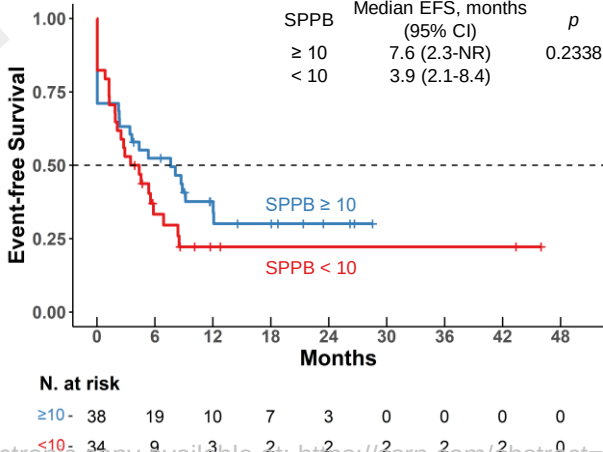
B. OS - HMA/VEN



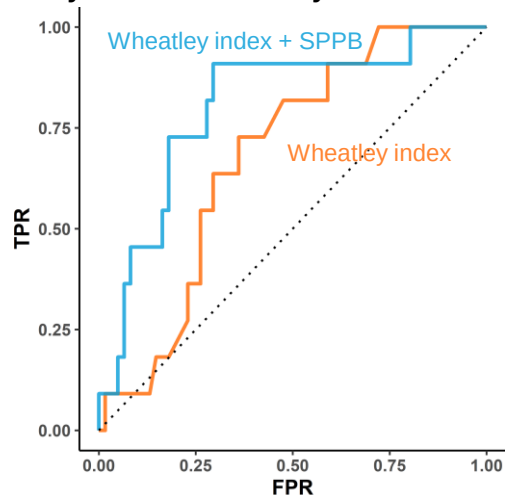
C. EFS - HMA



D. EFS - HMA/VEN



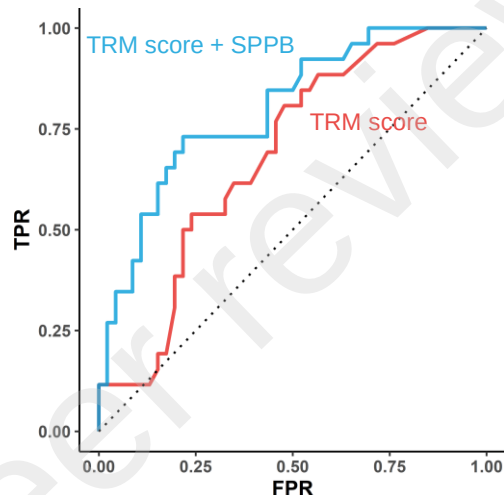
A. Wheatley index vs. Wheatley index + SPPB



Logistic regression model for early death

	Wheatley index	Wheatley index + SPPB	<i>p</i>
AUC	0.6803	0.8033	0.0306
Likelihood	AIC 61.59	AIC 57.92	0.0173
IDI		0.09	0.0240
Continuous NRI		1.05	<0.0001

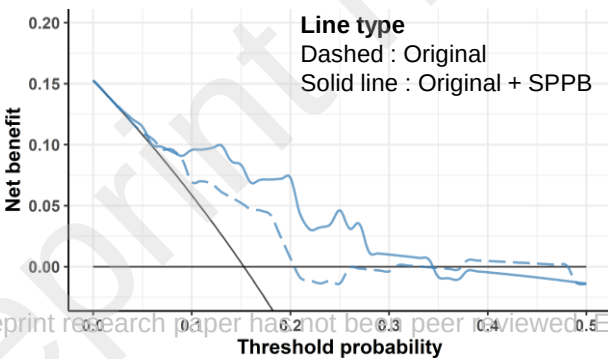
B. TRM score vs. TRM score + SPPB



Logistic regression model for early death

	TRM score	TRM score + SPPB	<i>p</i>
AUC	0.6835	0.7985	0.0113
Likelihood	AIC 89.25	AIC 78.75	0.0004
IDI		0.15	0.0009
Continuous NRI		0.95	<0.0001

C. Wheatley index vs. Wheatley index + SPPB



D. TRM score vs. TRM score + SPPB

